

Generative AI (GenAI) for DevOps and DevSecOps Engineers

To learn prompt engineering, using AI models for automation tasks, and practical GenAI tools in DevOps and DevSecOps.

You (CLI / App)



Python (LangChain)



Ollama Server (localhost:11434)



Local LLM (mistral / llama)

Check Ollama Version

```
PS C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS> ollama --version
ollama version is 0.13.5
```

Step 3: Test a Model

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS> & C:/Users/Amit/Desktop/Int_Prep/GITHUB_PROJECTS/.venv/Scripts/Activate.ps1
(.venv) PS C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS> ollama --version
ollama version is 0.13.5
(.venv) PS C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS> ollama pull mistral
pulling manifest
pulling f5074b1221da: 100% 4.4 GB
pulling 43070e2d4e53: 100% 11 KB
pulling 1ff5b64b61b9: 100% 799 B
pulling ed11eda7790d: 100% 30 B
pulling 1064e17101bd: 100% 487 B
verifying sha256 digest
writing manifest
success
(.venv) PS C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS> ollama run mistral
>>> send a message (? for help)
```

```

PS C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS> ollama serve
time=2025-12-29T20:43:47.215+05:30 level=INFO source=routes.go:1554 msg="server config" env="map[CUDA_VISIBLE_DEVICES: GGML_VK_VISIBLE_DEVICES: GPU_DEVICE_ORDINAL:
HTTP_VISIBLE_DEVICES: HSA_OVERRIDE_GFX_VERSION: HTTPS_PROXY: HTTP_PROXY: NO_PROXY: OLLAMA_CONTEXT_LENGTH:4096 OLLAMA_DEBUG:INFO OLLAMA_FLASH_ATTENTION:false OLLAMA
GPU_OVERHEAD:0 OLLAMA_HOST:http://127.0.0.1:11434 OLLAMA_KEEP_ALIVE:5m0s OLLAMA_KV_CACHE_TYPE: OLLAMA_LLM_LIBRARY: OLLAMA_LOAD_TIMEOUT:5m0s OLLAMA_MAX_LOADED_MODE
LS:0 OLLAMA_MAX_QUEUE:512 OLLAMA_MODELS:C:\Users\Amit\ollama\models OLLAMA_MULTIUSER_CACHE:false OLLAMA_NEW_ENGINE:false OLLAMA_NOHISTORY:false OLLAMA_NOPRUNE
:false OLLAMA_NUM_PARALLEL:1 OLLAMA_ORIGINS:[http://localhost https://localhost http://localhost:* https://localhost:* http://127.0.0.1 https://127.0.0.1 http://12
7.0.0.1:* https://127.0.0.1:* http://0.0.0.0 https://0.0.0.0 http://0.0.0.0:* https://0.0.0.0:* app://* file://* tauri://* vscode-webview://* vscode-file://*] OLLA
MA_REMOTES:[ollama.com] OLLAMA_SCHED_SPREAD:false OLLAMA_VULKAN:false ROCR_VISIBLE_DEVICES:]"
time=2025-12-29T20:43:47.232+05:30 level=INFO source=images.go:493 msg="total blobs: 10"
time=2025-12-29T20:43:47.234+05:30 level=INFO source=images.go:500 msg="total unused blobs removed: 0"
time=2025-12-29T20:43:47.244+05:30 level=INFO source=routes.go:1607 msg="listening on 127.0.0.1:11434 (version 0.13.5)"
time=2025-12-29T20:43:47.261+05:30 level=INFO source=runner.go:67 msg="discovering available GPUs..."
time=2025-12-29T20:43:47.358+05:30 level=INFO source=server.go:429 msg="starting runner" cmd="C:\\Users\\Amit\\AppData\\Local\\Programs\\Ollama\\ollama.exe runner
--ollama-engine --port 53315"
time=2025-12-29T20:43:47.673+05:30 level=INFO source=server.go:429 msg="starting runner" cmd="C:\\Users\\Amit\\AppData\\Local\\Programs\\Ollama\\ollama.exe runner
--ollama-engine --port 53327"
time=2025-12-29T20:43:49.314+05:30 level=INFO source=server.go:429 msg="starting runner" cmd="C:\\Users\\Amit\\AppData\\Local\\Programs\\Ollama\\ollama.exe runner
--ollama-engine --port 53339"
time=2025-12-29T20:43:49.882+05:30 level=INFO source=runner.go:106 msg="experimental Vulkan support disabled. To enable, set OLLAMA_VULKAN=1"
time=2025-12-29T20:43:49.884+05:30 level=INFO source=types.go:60 msg="inference compute" id=cpu library=cpu compute="" name=cpu description=cpu libdirs=ollama driv
ers="" pci_id="" type="" total="15.9 GiB" available="7.6 GiB"
time=2025-12-29T20:43:49.892+05:30 level=INFO source=routes.go:1648 msg="entering low vram mode" "total vram"="0 B" threshold="20.0 GiB"

```

Check port:

```

PS C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS> netstat -ano | findstr 11434
TCP        0.0.0.0:11434          0.0.0.0:0             LISTENING   16004
TCP        [::]:11434            [::]:0                 LISTENING   16004
PS C:\Users\Amit\Desktop\Int_Prep\GITHUB_PROJECTS> history

```